

AHMAD NAUFAL FARRAS

Surabaya, Indonesia

☎ +62 813-3138-3599 ✉ ahmadnaufalfarras@gmail.com  [linkedin.com/in/ahmad-naufal-farras-182327340](https://www.linkedin.com/in/ahmad-naufal-farras-182327340)  github.com/eycoo

Education

Institut Teknologi Sepuluh Nopember

Jul. 2024 – Present

Bachelor of Science in Artificial Intelligence Engineering (GPA: 3.4/4.00)

Surabaya, Indonesia

Experience

MengAsihi RAG Based AI Chatbot

May. 2026 – Present

Freelance AI Engineer

Surabaya, Indonesia

- Designing a Retrieval-Augmented Generation (RAG) pipeline for a breastfeeding guidance chatbot, with tech stack selection covering LLM, vector database, and retrieval strategy for a sensitive health domain.
- Architecting a microservice backend with REST API and native Android frontend, with focus on low inference latency and response guardrails to mitigate hallucination risks.

Avalon AI Corporation

Apr. 2026 – Present

AI Research Assistant

Surabaya, Indonesia

- Conducting proprietary foundational model research under NDA, focusing on neuromorphic vision and spike-based image classification using Izhikevich neuron dynamics and temporal encoding schemes.
- Implemented a custom SNN training pipeline from scratch in pure PyTorch including custom layers, surrogate gradient functions, and temporal encoders without reliance on existing SNN frameworks.

Schematics

Feb. 2026 – Present

Expert Staff of Data Science Sub Division

Surabaya, Indonesia

- Pioneered the technical foundation of Schematics' newly established Data Science competition division, contributing as one of the first expert staff to define its structure and standards.
- Conducted technical review and validation of competition datasets to ensure data quality, integrity, and suitability for competitive use.

Innovillage

Jan. 2026 – Present

AI Engineer

Sidoarjo, Indonesia

- Secured funding to develop a digital village solution focused on optimizing Tilapia farming.
- Engineered a Computer Vision model using Custom CNNs with CBAM attention modules to detect fish diseases and classify health status.

Pusat Studi Kecerdasan Artifisial dan Teknologi Digital

Jan. 2026 – Feb. 2026

AI Research Intern

Surabaya, Indonesia

- Developed an EEG signal analysis pipeline to classify ALS vs. Normal subjects across 1,809 EDF recordings (6 ALS, 150 Normal subjects) with 9 task scenarios.
- Engineered a chunking and chain encoding pipeline extracting time-domain features (MAV, Variance, STD) per subband (Delta, Theta, Alpha, Beta, Gamma) from segmented EEG tasks (Resting, Thinking, Typing).

REXTRA

August. 2025 – Feb. 2026

AI Engineer

Surabaya, Indonesia

- Maintained and optimized REST API integration for LLM inference calls, ensuring reliable communication between the backend service and external model providers.
- Monitored API response handling, error management, and latency.

ITS Challenge

Jun. 2025 – Sept. 2025

Staff of Data Center

Surabaya, Indonesia

- Managed the collection, validation, and integrity of participant databases for university events.
- Optimized data workflows using spreadsheet to ensure efficient information retrieval.

INCLENATION

Jun. 2025 – Aug. 2025

Staff of Data Management

Surabaya, Indonesia

- Managed the collection, validation, and integrity of participant databases for university events.
- Optimized data workflows using spreadsheet to ensure efficient information retrieval.

Projects

Medical Chest X-Ray Classification via Stable Diffusion | *Computer Vision* 2026

- Classified chest X-ray images into 6 categories by utilizing a frozen Stable Diffusion U-Net as a feature extractor, integrated with a Dual Feature Aggregation module.
- Outperformed DINOv2, MaxViT, and ConvNeXtV2 baselines across 4 experimental scenarios; achieved best F1-Macro of 45.17% and AUC-OVR of 77.47% on balanced data with feature-space augmentation, and 38.55% F1-Macro on imbalanced setting without augmentation.

Chat to Core | *NLP, Computer Vision, System Design* 2026

- Led a 4-person team to build a WhatsApp based AI system converting unstructured UMKM transaction data into SAK EMKM compliant financial reports, credit scores, and RAG powered advisory.
- Designed a microservice architecture integrating IndoBERT for NER, Whisper for ASR, YOLOv8 Nano and PaddleOCR for receipt parsing, and Phi-3 Mini with pgvector for retrieval augmented generation.

Synthetic Data for Indonesian Ethnicities | *Stable Diffusion XL, LoRA* 2025

- Engineered a data generation pipeline using **Stable Diffusion XL** to resolve severe class imbalance in Indonesian tribal datasets.
- Fine-tuned the model using **Low-Rank Adaptation (LoRA)** to synthesize high-fidelity images that preserve distinct ethnic facial features. +
- Demonstrated that augmenting the training set improved external classifier (XGBoost/ViT) accuracy, proving the efficacy of synthetic data in low-resource scenarios.

Sequential vs Frequency Approach Analysis on Audio Classification | *HMM, Naive Bayes, Audio Processing* 2025

- Analyzed the effectiveness of sequential (HMM) vs frequency-based (Multinomial Naive Bayes) approaches for audio sound classification using MFCC feature extraction and Vector Quantization via K-Means clustering.
- Evaluated both models across 3 scenarios: codebook size sensitivity, training data size impact, and audio duration robustness, with Naive Bayes achieving up to 90% accuracy and 95% precision at full training data.
- Demonstrated that Naive Bayes consistently outperformed HMM in frequency-dominant tasks, while HMM showed competitive performance at smaller codebook sizes (85% accuracy at 32 clusters).

TB Cough Detection System | *Deep Learning, LSTM, Audio Processing* 2025

- Winner of Gold Medal at Software Development Division Gemastik XVIII:** Developed a bio-acoustic diagnosis system to detect Tuberculosis from cough sounds.
- Implemented a feature extraction pipeline using **MFCCs** and spectral features to capture distinct respiratory patterns.
- Designed and trained a **Long Short-Term Memory (LSTM)** network optimized for sequential audio data, achieving high sensitivity in distinguishing TB coughs from healthy subjects.

CodeSight | *Computer Vision, YOLOv8, OpenCV* 2025

- Built an accessible learning tool for visually impaired students that translates physical flowchart blocks into executable code via Computer Vision.
- Deployed a custom-trained **YOLO model** for real-time object detection, handling variable lighting and block orientations.
- Enabled students to learn programming logic independently by providing immediate audio feedback on their physical code arrangements.

Cough Pose Detection & DoA | *Computer Vision, Audio Processing* 2025

- Built a validation system combining Direction of Arrival (DoA) audio tracking with visual Pose Detection.
- Designed the algorithm to pinpoint and verify the specific individual coughing within a crowded environment.

Honors & Awards

Gold Medal, Software Development Division - Gemastik (2025)

Top 50, Social Project Program Innovillage (2025)

1st Winner, BRIN AIDeaNation (2025)

Bronze Medal, Youth Scientific Work World Innovative Science Project Olympiad (WISPO) (2023)

Technical Skills

Languages: Python, SQL, C++

Frameworks/Libraries: PyTorch, TensorFlow, Scikit-Learn, Pandas, OpenCV, HuggingFace Transformers, LangChain, FastAPI

Developer Tools: Git, VS Code, Google Colab, Kaggle, REST API

Soft Skills: Problem Solving, Teamwork, Time Management